

HPCC AWS Cluster



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HPCC AWS Cluster

HPCC

Your HPCC account,
HPCC's configurations.



AWS (Amazon Web Services)

Your AWS account, AWS's
massive compute
resources.



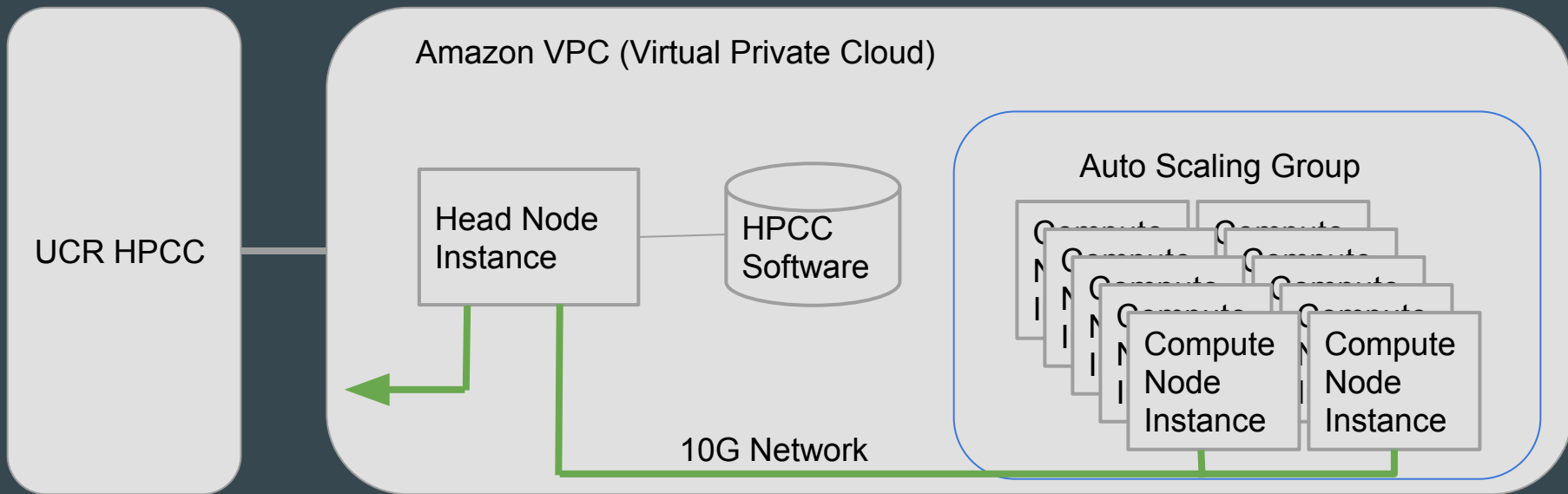
HPCC AWS Cluster

Using HPCC's configurations and AWS's
resources together; users can create complete
private clusters simply and quickly providing on
demand cluster computing.

HPCC AWS Cluster

What is it?

It's a Real Cluster!



HPCC AWS Cluster

Through AWS tools and custom HPCC configurations we enable users to:

- Build a private cluster in ~15 min.
 - Any number of nodes, auto scaling (up and down), limit 20 to start
 - Any type of compute nodes
 - High memory
 - High CPU
 - GPUs
 - Choose HPCC base type
 - For only as long as you need it
 - Familiar interface and job scheduler
 - Easy ability to have the software you need installed by HPCC, if it's not there already
- Build as many of these clusters as you need (even at the same time)
- Pay for only the time you use it - per/min billing

Setup

Overview

1

Create AWS Account

Create a single account for you or create accounts for your lab. Setup for invoicing and AWS/UCR Data Egress Waiver.

2

Configure your cluster

Login to your HPCC account and run configure script.

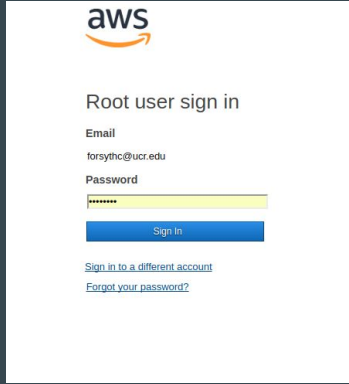
3

Create and use cluster

Create a cluster with one command and 15 min.. Once built, login, upload data, compute, download results, and delete the cluster.

Setup - Create AWS Account

Login



The image shows the AWS root user sign-in page. At the top is the AWS logo. Below it is the text "Root user sign in". There is a field for "Email" with the value "forsythc@ucr.edu" and a field for "Password" with masked characters. A blue "Sign In" button is below the password field. At the bottom, there are two links: "Sign in to a different account" and "Forgot your password?".

aws

Root user sign in

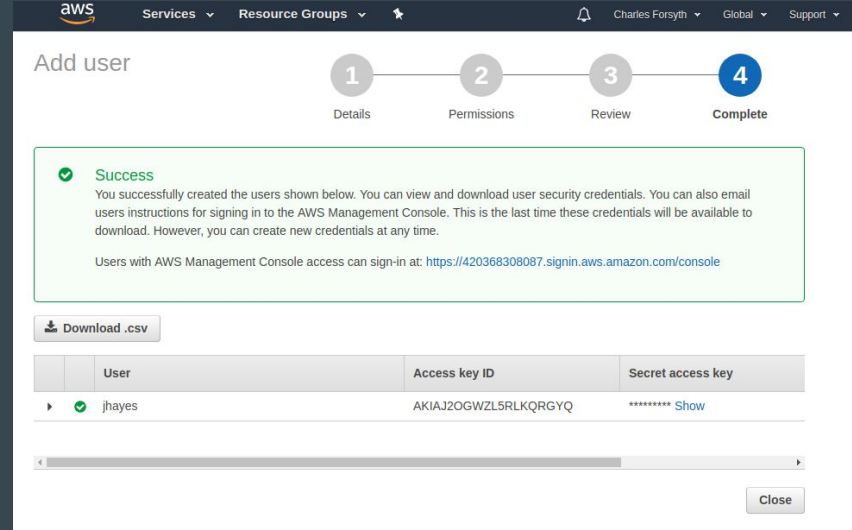
Email
forsythc@ucr.edu

Password

Sign In

[Sign in to a different account](#)
[Forgot your password?](#)

Create AWS Account



The image shows the "Add user" success screen in the AWS Management Console. At the top, there is a progress bar with four steps: 1. Details, 2. Permissions, 3. Review, and 4. Complete (highlighted in blue). Below the progress bar is a green success message box that says "Success" and provides instructions on how to sign in. Below the message box is a "Download .csv" button. At the bottom, there is a table with user details.

aws Services Resource Groups

Add user

1 Details 2 Permissions 3 Review 4 Complete

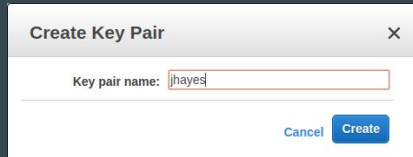
Success
You successfully created the users shown below. You can view and download user security credentials. You can also email users instructions for signing in to the AWS Management Console. This is the last time these credentials will be available to download. However, you can create new credentials at any time.
Users with AWS Management Console access can sign-in at: <https://420368308087.signin.aws.amazon.com/console>

Download .csv

	User	Access key ID	Secret access key
▶	✓ jhayes	AKIAJ2OGWZL5RLKQRGYQ	***** Show

Close

Create Key Pair



The image shows the "Create Key Pair" dialog box. It has a title bar with "Create Key Pair" and a close button. Inside, there is a label "Key pair name:" followed by a text input field containing the value "jhayes". At the bottom, there are two buttons: "Cancel" and "Create".

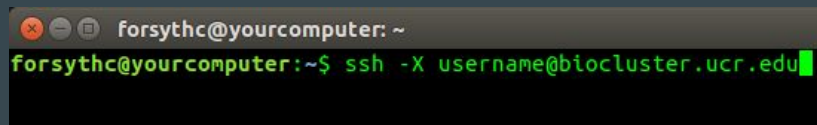
Create Key Pair

Key pair name: jhayes

Cancel Create

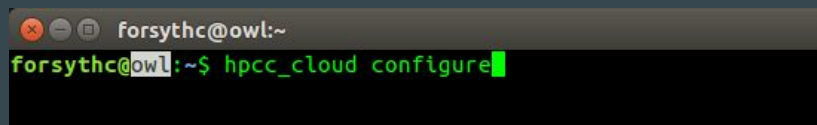
Setup - Configure HPCC

Connect to HPCC

A terminal window with a dark background. The title bar shows 'forsythc@yourcomputer: ~'. The prompt is 'forsythc@yourcomputer:~\$' and the command 'ssh -X username@biocluster.ucr.edu' is entered, followed by a green cursor.

```
forsythc@yourcomputer: ~  
forsythc@yourcomputer:~$ ssh -X username@biocluster.ucr.edu
```

hpcc_cloud configure

A terminal window with a dark background. The title bar shows 'forsythc@owl:~'. The prompt is 'forsythc@owl:~\$' and the command 'hpcc_cloud configure' is entered, followed by a green cursor.

```
forsythc@owl:~  
forsythc@owl:~$ hpcc_cloud configure
```

Cluster Operations

Cluster Operations - Overview



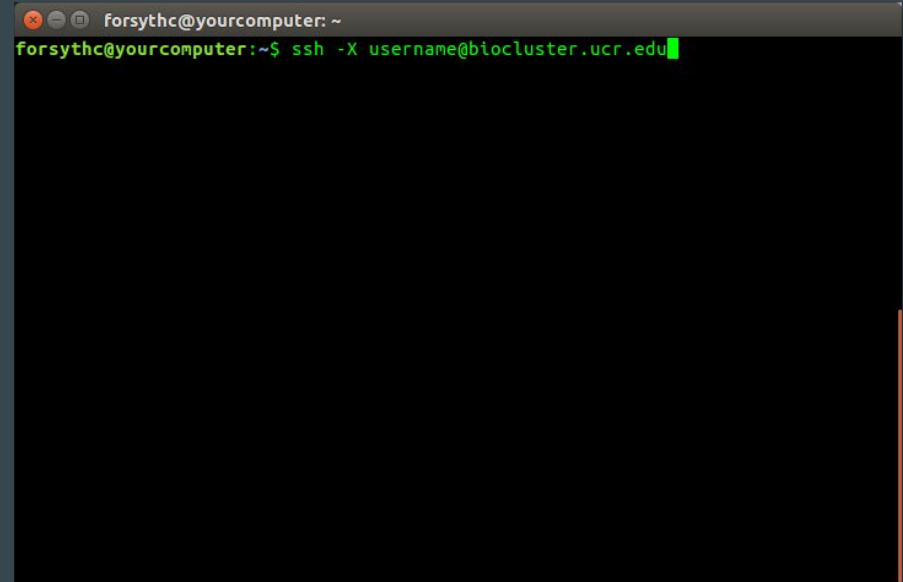
Cluster Operations - Overview

hpcc_cloud

- hpcc_cloud list
- hpcc_cloud create <NameForYourCluster>
- hpcc_cloud status <NameForYourCluster>
- hpcc_cloud delete <NameForYourCluster>
- hpcc_cloud nodetype
- Connecting to your cluster
 - ssh -i /path/to/your/key-file.pem centos@<MasterPublicIP>
- Uploading data to your cluster
 - scp -i /path/to/your/key-file.pem <local-files-to-copy> centos@<MasterPublicIP>:.
- Downloading data/results
 - scp -i /path/to/your/key-file.pem centos@<MasterPublicIP>:./<files-to-download> .

Cluster Operations - Connect to HPCC

Connect to HPCC



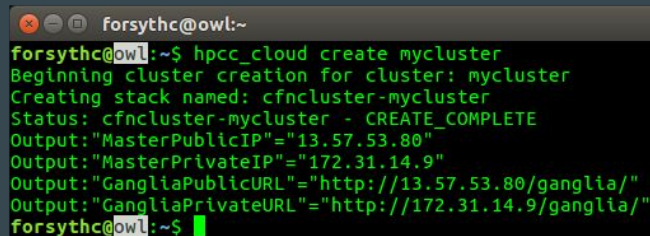
```
forsythc@yourcomputer: ~  
forsythc@yourcomputer:~$ ssh -X username@biocluster.ucr.edu
```

A terminal window with a dark background. The title bar shows window controls and the text 'forsythc@yourcomputer: ~'. The prompt 'forsythc@yourcomputer:~\$' is followed by the command 'ssh -X username@biocluster.ucr.edu' and a green cursor.

Cluster Operations - Create a Cluster

`hpcc_cloud create <NameForYourCluster>`

- `<NameForYourCluster>` can be anything you like.
- cluster build process ~15 min.

A terminal window titled 'forsythc@owl:~' showing the execution of the 'hpcc_cloud create mycluster' command. The output shows the cluster creation process, including stack creation and completion status, along with public and private IP addresses and Ganglia URLs.

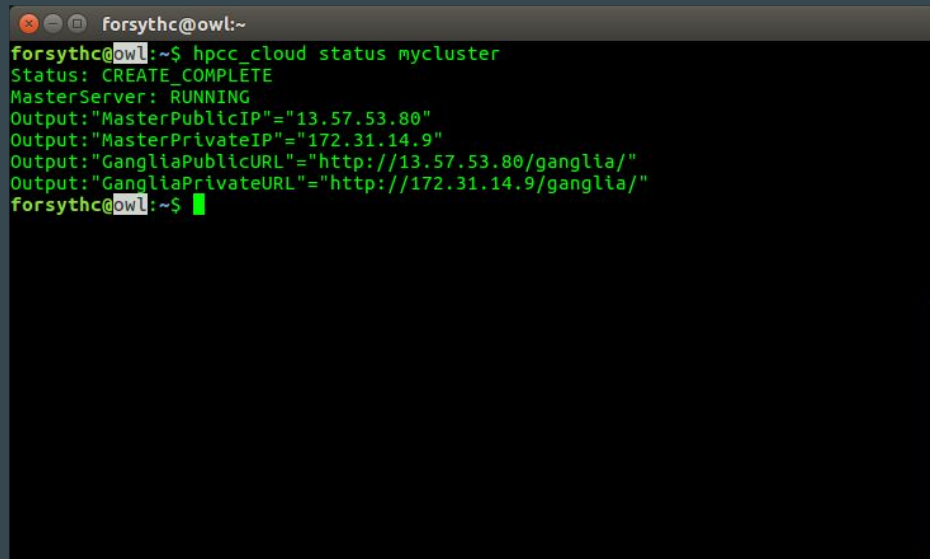
```
forsythc@owl:~$ hpcc_cloud create mycluster
Beginning cluster creation for cluster: mycluster
Creating stack named: cfncluster-mycluster
Status: cfncluster-mycluster - CREATE_COMPLETE
Output: "MasterPublicIP"="13.57.53.80"
Output: "MasterPrivateIP"="172.31.14.9"
Output: "GangliaPublicURL"="http://13.57.53.80/ganglia/"
Output: "GangliaPrivateURL"="http://172.31.14.9/ganglia/"
forsythc@owl:~$
```

Cluster Operations - Cluster Status

hpcc_cloud status

<NameForYourCluster>

- MasterPublicIP is used to connect the cluster as well as upload and download files.
- GangliaPublicURL is a web based cluster monitoring tool.

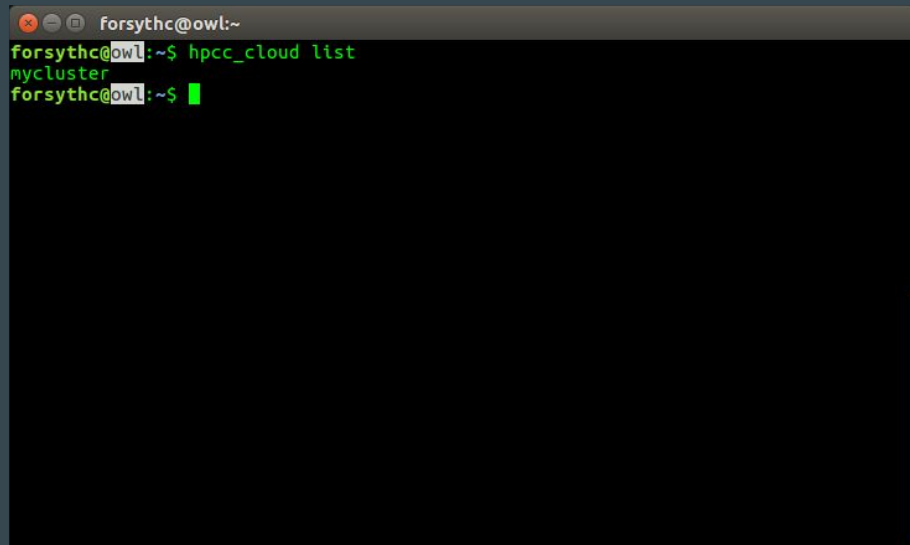
A terminal window with a dark background and light green text. The window title is 'forsythc@owl:~'. The command 'hpcc_cloud status mycluster' has been executed, resulting in the following output: 'Status: CREATE_COMPLETE', 'MasterServer: RUNNING', 'Output: "MasterPublicIP"="13.57.53.80"', 'Output: "MasterPrivateIP"="172.31.14.9"', 'Output: "GangliaPublicURL"="http://13.57.53.80/ganglia/"', and 'Output: "GangliaPrivateURL"="http://172.31.14.9/ganglia/"'. The prompt 'forsythc@owl:~\$' is visible at the bottom.

```
forsythc@owl:~$ hpcc_cloud status mycluster
Status: CREATE_COMPLETE
MasterServer: RUNNING
Output: "MasterPublicIP"="13.57.53.80"
Output: "MasterPrivateIP"="172.31.14.9"
Output: "GangliaPublicURL"="http://13.57.53.80/ganglia/"
Output: "GangliaPrivateURL"="http://172.31.14.9/ganglia/"
forsythc@owl:~$
```

Cluster Operations - List Active Clusters

hpcc_cloud list

- Output = list of running clusters

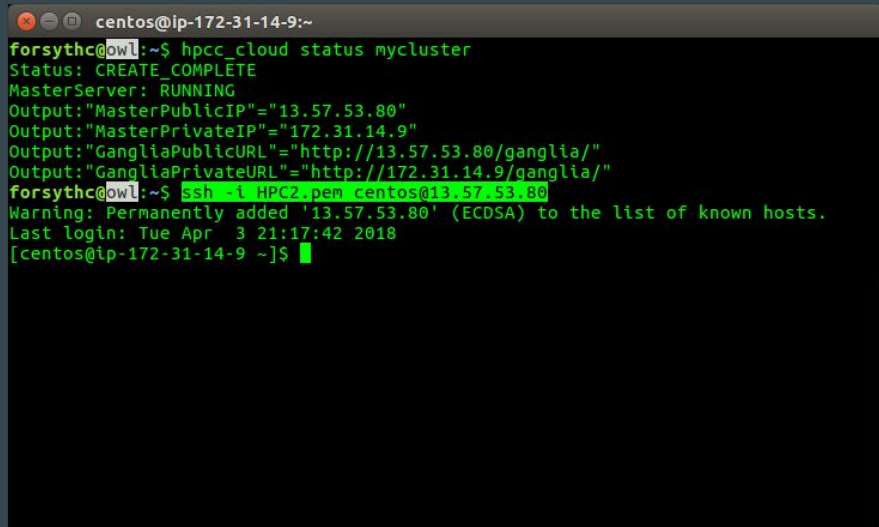


```
forsythc@owl:~$ hpcc_cloud list
mycluster
forsythc@owl:~$
```

Cluster Operations - Connect to your Cluster

```
ssh -i /path/to/your/key-file.pem  
centos@<MasterPublicIP>
```

- /path/to/your/key-file.pem
 - place where you saved your key-file.pem
- <MasterPublicIP>
 - This is the public ip address of your cluster



```
centos@ip-172-31-14-9:~  
forsythc@owl:~$ hpcc_cloud status mycluster  
Status: CREATE_COMPLETE  
MasterServer: RUNNING  
Output: "MasterPublicIP"="13.57.53.80"  
Output: "MasterPrivateIP"="172.31.14.9"  
Output: "GangliaPublicURL"="http://13.57.53.80/ganglia/"  
Output: "GangliaPrivateURL"="http://172.31.14.9/ganglia/"  
forsythc@owl:~$ ssh -i HPC2.pem centos@13.57.53.80  
Warning: Permanently added '13.57.53.80' (ECDSA) to the list of known hosts.  
Last login: Tue Apr 3 21:17:42 2018  
[centos@ip-172-31-14-9 ~]$
```

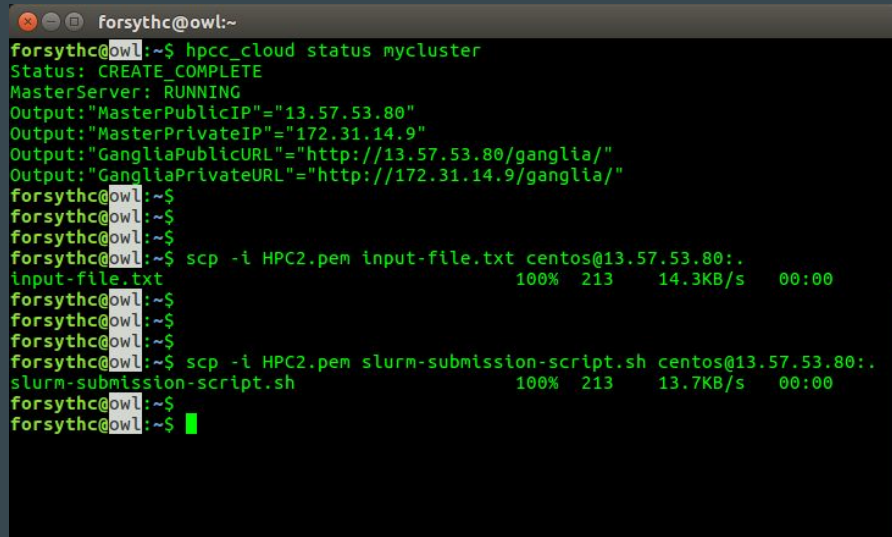
Cluster Operations - Upload to your Cluster

```
scp -i /path/to/your/key-file.pem
```

```
<local-files-to-copy>
```

```
centos@<MasterPublicIP>:.
```

- /path/to/your/key-file.pem
 - place where you saved your key-file.pem
- <MasterPublicIP>
 - replace with MasterPublicIP
- <local-files-to-copy>
 - path to the files you wish to upload to your cluster

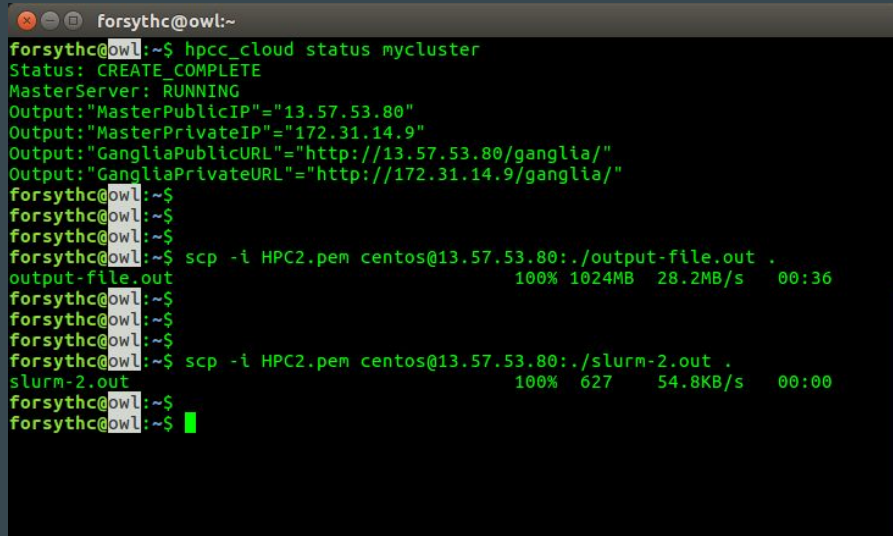
A terminal window titled 'forsythc@owl:~' showing the execution of 'hpcc_cloud status mycluster'. The output indicates the cluster is in 'CREATE_COMPLETE' status with the master server running. It lists the MasterPublicIP as 13.57.53.80 and MasterPrivateIP as 172.31.14.9, along with Ganglia URLs. Following this, two 'scp' commands are used to upload 'input-file.txt' and 'slurm-submission-script.sh' to the master node. Both uploads complete successfully at 100% speed.

```
forsythc@owl:~$ hpcc_cloud status mycluster
Status: CREATE_COMPLETE
MasterServer: RUNNING
Output: "MasterPublicIP"="13.57.53.80"
Output: "MasterPrivateIP"="172.31.14.9"
Output: "GangliaPublicURL"="http://13.57.53.80/ganglia/"
Output: "GangliaPrivateURL"="http://172.31.14.9/ganglia/"
forsythc@owl:~$
forsythc@owl:~$
forsythc@owl:~$
forsythc@owl:~$ scp -i HPC2.pem input-file.txt centos@13.57.53.80:.
input-file.txt                                100% 213    14.3KB/s   00:00
forsythc@owl:~$
forsythc@owl:~$
forsythc@owl:~$
forsythc@owl:~$ scp -i HPC2.pem slurm-submission-script.sh centos@13.57.53.80:.
slurm-submission-script.sh                    100% 213    13.7KB/s   00:00
forsythc@owl:~$
forsythc@owl:~$
```

Cluster Operations - Download from your Cluster

```
scp -i /path/to/your/key-file.pem  
centos@<MasterPublicIP>:./<files-to-downl  
oad> .
```

- /path/to/your/key-file.pem
 - place where you saved your key-file.pem
- <MasterPublicIP>
 - replace with MasterPublicIP
- <files-to-download>
 - files you wish to download from your cluster



```
forsythc@owl:~  
forsythc@owl:~$ hpcc_cloud status mycluster  
Status: CREATE_COMPLETE  
MasterServer: RUNNING  
Output:"MasterPublicIP"="13.57.53.80"  
Output:"MasterPrivateIP"="172.31.14.9"  
Output:"GangliaPublicURL"="http://13.57.53.80/ganglia/"  
Output:"GangliaPrivateURL"="http://172.31.14.9/ganglia/"  
forsythc@owl:~$  
forsythc@owl:~$  
forsythc@owl:~$  
forsythc@owl:~$ scp -i HPC2.pem centos@13.57.53.80:./output-file.out .  
output-file.out                                100% 1024MB 28.2MB/s 00:36  
forsythc@owl:~$  
forsythc@owl:~$  
forsythc@owl:~$  
forsythc@owl:~$ scp -i HPC2.pem centos@13.57.53.80:./slurm-2.out .  
slurm-2.out                                     100% 627 54.8KB/s 00:00  
forsythc@owl:~$  
forsythc@owl:~$
```

Cluster Operations - Example

Simple Job Submission Example

1. Create cluster
2. Upload files
3. Login
4. Submit Job
5. Check queue
6. Exit Cluster
7. Download output file
8. Delete Cluster

```
forsythc@ivxx:~$ ssh forsythc@biocluster.ucr.edu
Last login: Thu Mar 22 11:42:04 2018 from 138.23.58.54

-----
University of California, Riverside - HPCC (High-Performance Computing Center)
-----

More information about HPCC and how to use the resources provided can
be found at http://hpcc.ucr.edu/manuals/linux-cluster\_intro.html

Please send all questions and support requests to support@biocluster.ucr.edu

Note: The default version of R is now 3.4.2
-----

forsythc@pigeon:~$ hpcc cloud create example
Beginning cluster creation for cluster: example
Creating stack named: cfncluster-example
Status: cfncluster-example - CREATE COMPLETE
Output: "MasterPrivateIP"="172.31.6.68"
Output: "GangliaPublicURL"="http://13.57.1.17/ganglia/"
Output: "GangliaPrivateURL"="http://172.31.6.68/ganglia/"
forsythc@pigeon:~$ ssh -i ~/AWS/Keys/HPC2.pem centos@13.57.1.17
Warning: Permanently added '13.57.1.17' (ECDSA) to the list of known hosts.
Last login: Thu Mar 22 19:10:51 2018
[centos@ip-172-31-6-68 ~]$ sinfo
PARTITION AVAIL  TIMELIMIT  NODES  STATE NODELIST
compute*    up       infinite    1    idle ip-172-31-13-177
[centos@ip-172-31-6-68 ~]$ exit
logout
Connection to 13.57.1.17 closed.
forsythc@pigeon:~$ cd hpcc_cloud-example/
forsythc@pigeon:~/hpcc_cloud-example$ scp -i ~/AWS/Keys/HPC2.pem printenv.sh centos@13.57.1.17:.
```



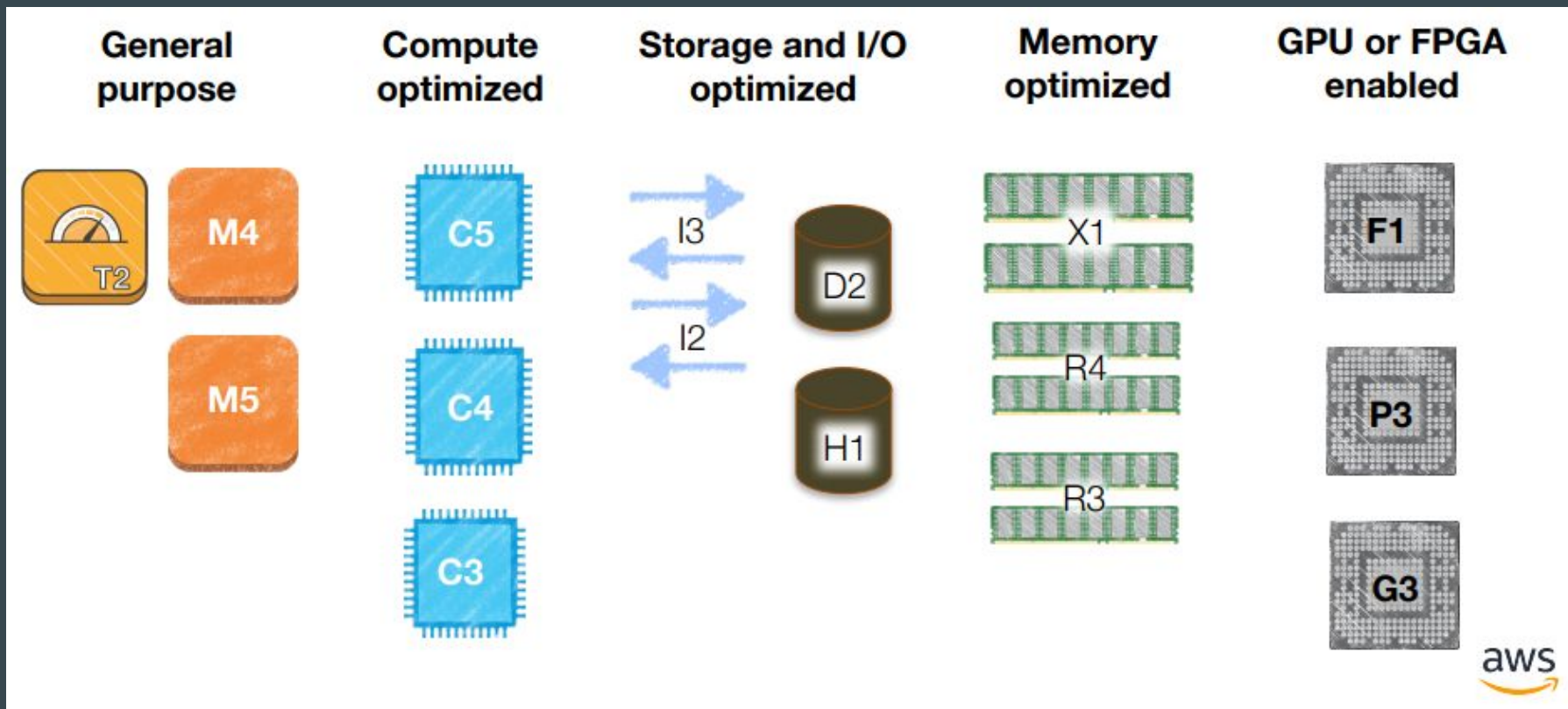
Cluster Operations - Nodetypes

hpcc_cloud nodetype

- hpcc_cloud nodetype
- hpcc_cloud nodetype avail
- hpcc_cloud nodetype test
- hpcc_cloud nodetype base
- hpcc_cloud nodetype gpu

```
forsythc@owl:~$ hpcc_cloud nodetype
Current node type is c5.2xlarge (@ $0.34 per hour per node while cluster is running)
forsythc@owl:~$ hpcc_cloud nodetype avail
Node type: t2.micro      Keyword: test      Cost: 0.012
Node type: c5.2xlarge   Keyword: base      Cost: 0.340
Node type: p2.8xlarge   Keyword: gpu       Cost: 7.200
Node type: r4.large     Keyword: highmem   Cost: 0.133
Node type: c5.9xlarge   Keyword: c5.9xlarge Cost: 1.530
Node type: x1.16xlarge  Keyword: x1.16xlarge Cost: 6.669
Node type: i3.2xlarge   Keyword: i3.2xlarge Cost: 0.624
Node type: m5.large     Keyword: m5.large  Cost: 0.096
forsythc@owl:~$ hpcc_cloud nodetype test
Cluster node type set to test - t2.micro (@ $0.0116 per hour per node while cluster is running)
forsythc@owl:~$ hpcc_cloud nodetype base
Cluster node type set to base - c5.2xlarge (@ $0.34 per hour per node while cluster is running)
forsythc@owl:~$ hpcc_cloud nodetype gpu
Cluster node type set to gpu - p2.8xlarge (@ $7.2 per hour per node while cluster is running)
forsythc@owl:~$
```

Cluster Operations - Nodetypes



Billing and Cost Control

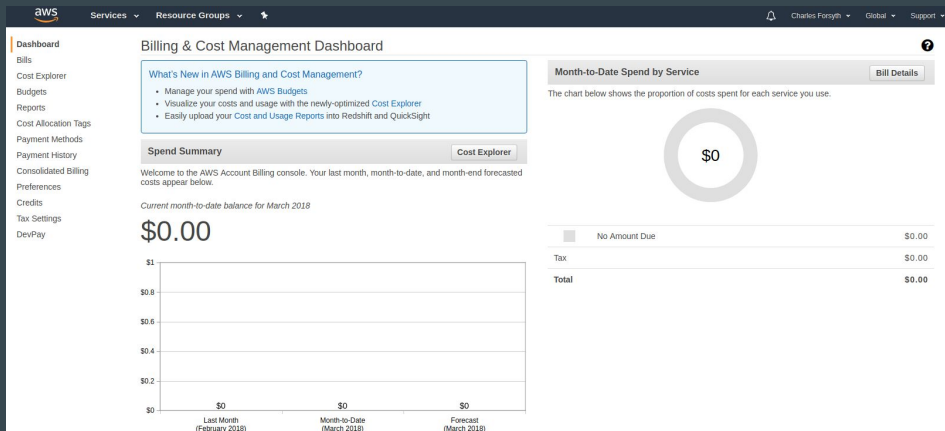
Costs and Billing

- Users billed directly by Amazon through a PO associated with a FAU.
- Costs based on:
 - Data Transferred (UC/AWS Enterprise Agreement - egress waiver)
 - Data Storage
 - Infrastructure (VPC)
 - Compute and Head node types used
 - These vary based on job requirements
 - High memory
 - High CPU single/multi-node
 - GPUs
 - Can use spot pricing to lower costs in some cases
- Budget alerts and built in limits can be used to monitor and control spending.

Node Types and Approximate Costs

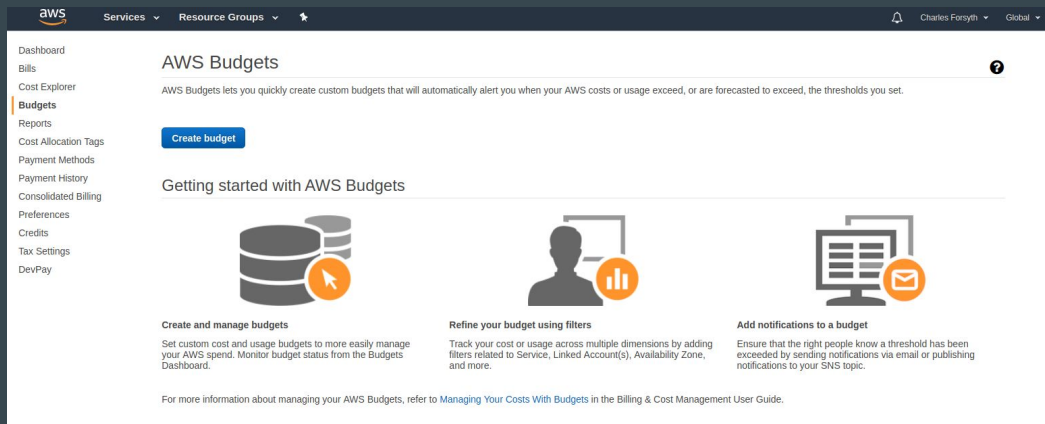
Description	nodes	Usage	Type	Cost/hour	Total Cost/hour
Low Performance 1 Core	1	1	t2.micro	0.02	0.02
16 Core	1	1	c3.4xlarge	0.84	0.84
32 Core	1	1	c3.8xlarge	1.68	1.68
64 Core	1	1	m4.16xlarge	3.2	3.2
32 Core x 32 = 1024 Core	32	1	c3.8xlarge	1.68	53.76
GPU with 1 GPUs	1	1	p2.xlarge	0.9	0.9
GPU with 8 GPUs	1	1	p2.8xlarge	7.2	7.2
High Mem 64 Cores	1	1	x1.16xlarge	6.67	6.67
High Mem 128 Cores	1	1	x1.32xlarge	13.34	13.34
S3 storage	First 50 TB / month		\$0.026 per GB		
Amazon EBS General Purpose SSD (gp2) volumes			\$0.12 per GB-month of provisioned storage		

Cost Control - Budgets and Alerts



AWS Billing & Cost Management Dashboard

Cost Control - Budgets and Alerts



The screenshot shows the AWS Budgets console. At the top, there's a navigation bar with the AWS logo, 'Services', 'Resource Groups', and a user profile 'Charles Forsyth'. A left-hand sidebar lists navigation options: Dashboard, Bills, Cost Explorer, Budgets (highlighted), Reports, Cost Allocation Tags, Payment Methods, Payment History, Consolidated Billing, Preferences, Credits, Tax Settings, and DevPay. The main content area is titled 'AWS Budgets' and includes a 'Create budget' button. Below this, a section 'Getting started with AWS Budgets' features three cards: 'Create and manage budgets' (with a database icon), 'Refine your budget using filters' (with a person and bar chart icon), and 'Add notifications to a budget' (with a monitor and envelope icon). Each card has a brief description of its function. At the bottom, a link points to the 'Managing Your Costs With Budgets' section of the Billing & Cost Management User Guide.

aws Services Resource Groups

Charles Forsyth Global

Dashboard
Bills
Cost Explorer
Budgets
Reports
Cost Allocation Tags
Payment Methods
Payment History
Consolidated Billing
Preferences
Credits
Tax Settings
DevPay

AWS Budgets

AWS Budgets lets you quickly create custom budgets that will automatically alert you when your AWS costs or usage exceed, or are forecasted to exceed, the thresholds you set.

[Create budget](#)

Getting started with AWS Budgets



Create and manage budgets
Set custom cost and usage budgets to more easily manage your AWS spend. Monitor budget status from the Budgets Dashboard.



Refine your budget using filters
Track your cost or usage across multiple dimensions by adding filters related to Service, Linked Account(s), Availability Zone, and more.



Add notifications to a budget
Ensure that the right people know a threshold has been exceeded by sending notifications via email or publishing notifications to your SNS topic.

For more information about managing your AWS Budgets, refer to [Managing Your Costs With Budgets](#) in the Billing & Cost Management User Guide.

AWS Budgets

Cost Control - Budgets and Alerts

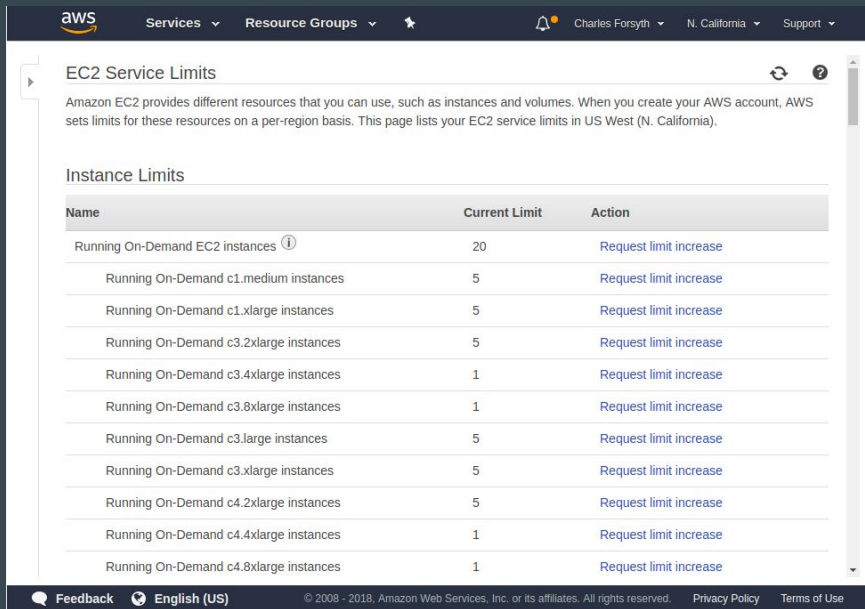
The screenshot shows the AWS Budgets console. On the left is a navigation menu with options like Dashboard, Bills, Cost Explorer, Budgets (highlighted), Reports, Cost Allocation Tags, Payment Methods, Payment History, Consolidated Billing, Preferences, Credits, Tax Settings, and DevPay. The main area is titled 'AWS Budgets' and includes buttons for 'Create budget', 'Copy', 'Edit', 'Delete', 'Download CSV', and a settings icon. Below these is a table of budgets.

		Budget type	Budget name	Current	Forecasted	Budgeted	Current vs. budgeted	Forecasted vs. budgeted
☐	▶	Cost	Monthly EC2 Budget	\$0.01	\$96.59	\$50.00	0%	193%

AWS budget created

- If we are forecasted to exceed the budget of \$50 we will get an email alert

Cost Control - Limits



The screenshot shows the AWS Management Console interface. At the top, the navigation bar includes the AWS logo, 'Services', 'Resource Groups', and user information for 'Charles Forsyth' in the 'N. California' region. The main content area is titled 'EC2 Service Limits' and includes a brief explanation: 'Amazon EC2 provides different resources that you can use, such as instances and volumes. When you create your AWS account, AWS sets limits for these resources on a per-region basis. This page lists your EC2 service limits in US West (N. California).' Below this, a section titled 'Instance Limits' contains a table with three columns: 'Name', 'Current Limit', and 'Action'.

Name	Current Limit	Action
Running On-Demand EC2 instances ⓘ	20	Request limit increase
Running On-Demand c1.medium instances	5	Request limit increase
Running On-Demand c1.xlarge instances	5	Request limit increase
Running On-Demand c3.2xlarge instances	5	Request limit increase
Running On-Demand c3.4xlarge instances	1	Request limit increase
Running On-Demand c3.8xlarge instances	1	Request limit increase
Running On-Demand c3.large instances	5	Request limit increase
Running On-Demand c3.xlarge instances	5	Request limit increase
Running On-Demand c4.2xlarge instances	5	Request limit increase
Running On-Demand c4.4xlarge instances	1	Request limit increase
Running On-Demand c4.8xlarge instances	1	Request limit increase

The footer of the console shows a 'Feedback' button, the language 'English (US)', and copyright information: '© 2008 - 2018, Amazon Web Services, Inc. or its affiliates. All rights reserved.' It also includes links for 'Privacy Policy' and 'Terms of Use'.

There is a limit to the number of each type of node you can have in your cluster concurrently by default.

This is a control measure to stop someone from accidentally creating tens or hundreds or thousands of nodes.

These limits can be expanded with a simple request to AWS.

Additional Features

Features - Familiar Cluster Environment

The HPCC AWS Cluster has been designed to be very similar to the HPCC cluster.

- Same scheduler
 - Slurm
- Same software tree
 - modules
 - new software can be installed at request
- Same directory structure
 - \$bigdata for running work
 - \$home for job setup

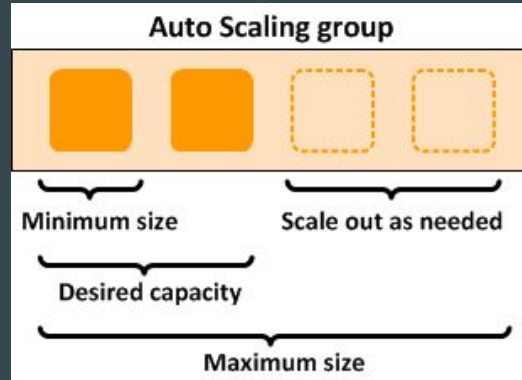
```
centos@ip-172-31-14-9:~  
Last login: Wed Apr  4 03:40:37 2018 from lb-3.bioinfo.ucr.edu  
[centos@ip-172-31-14-9 ~]$ ls -la  
total 40  
drwx-----. 4 centos centos    236 Apr  4 03:42 .  
drwxr-xr-x. 5 root root        49 Mar 20 18:52 ..  
-rw-----. 1 centos centos   4202 Apr  4 03:44 .bash_history  
-rw-r--r--. 1 centos centos    18 Dec  6 2016 .bash_logout  
-rw-r--r--. 1 centos centos   193 Dec  6 2016 .bash_profile  
-rw-r--r--. 1 centos centos   231 Dec  6 2016 .bashrc  
lrwxrwxrwx. 1 root root         8 Mar 20 18:52 bigdata -> /scratch  
-rw-r--r--. 1 centos centos    213 Apr  4 03:09 input-file.txt  
-rw-rw-r--. 1 centos centos 1073741824 Apr  4 03:15 output-file.out  
drwxrw----. 3 centos centos     19 Jan 11 23:51 .pki  
-rw-r--r--. 1 centos centos   3591 Apr  4 03:42 slurm-2.out  
-rw-r--r--. 1 centos centos    227 Apr  4 03:42 slurm-submission-script.sh  
drwx-----. 2 centos centos    104 Apr  3 21:17 .ssh  
-rw-----. 1 centos centos   4272 Apr  4 03:42 .viminfo  
[centos@ip-172-31-14-9 ~]$  
[centos@ip-172-31-14-9 ~]$  
[centos@ip-172-31-14-9 ~]$ module avail  
  
----- /usr/share/Modules/modulefiles -----  
dot          module-git  module-info  modules      null          use.own  
  
----- /etc/modulefiles -----  
mpi/mpich-3.0-x86_64  mpi/mpich-x86_64      mpi/openmpi-x86_64  
  
----- /opt/linux/centos/7.x/x86_64/modules -----  
bcftools/1.2(default)  java/8u45  
bowtie2/2.2.5(default) libgtextutils/0.6  
bwa/0.7.15(default)    miniconda2/4.4.10  
cfnccluster/1.4.1      pandoc/1.19.2.4  
fastqc/0.11.3(default) pandoc/2.0(default)  
fastx_toolkit/0.0.13(default) picard/1.130  
gatk/2.4-9(default)    picard/2.14.1(default)  
ggobi/2.1.11(default)  R/3.4.2(default)  
hisat2/2.1.0(default)  samtools/0.1.19  
java/6                  samtools/1.2  
java/6u43               samtools/1.3(default)  
java/7                  texlive/2017(default)  
java/7u40(default)      tophat/2.0.14(default)  
java/8                  trim_galore/0.4.2(default)  
java/8u25  
[centos@ip-172-31-14-9 ~]$
```

Features - Ganglia

- Auto created with each cluster
- Shows many cluster statistics
- URL/Link found in hpcc_cloud status command output



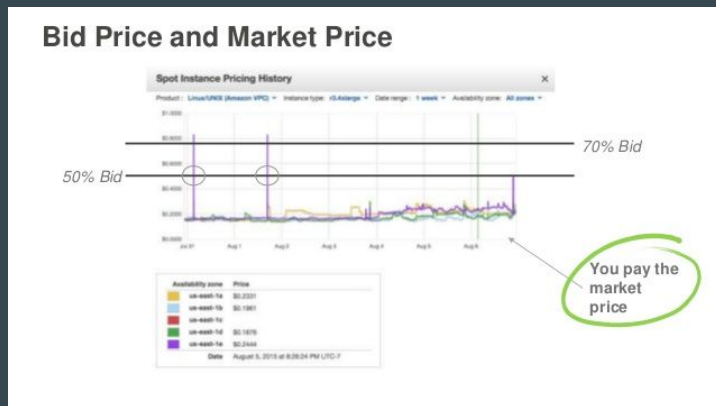
Features - Auto Scaling



- Always keep minimum number of instances running
- Never start more than maximum number of instances
- Launch or terminate instances to meet desired capacity
- Keeps costs at a minimum

Features - Spot Pricing

Spot Pricing



- AWS has a lot of computers and they don't like them to sit idle.
- They allow people to bid for this excess capacity at a very reduced rate; often 85% less.
- The downside is once that node is requested at full rate, the node will be shutdown and will be given to the requester that is paying full rates.
- *NOT very many HPC applications or workflows deal well with a node being turned off during a running job. While spot is available it will only work for specific work loads or job types.

Features - Configurable

One configuration file (stored in your home directory) controls the characteristics of the clusters you create.

Basic changes to the cluster configuration can be done through the `hpcc_cloud` command.

For advanced customization you are allowed to edit your configuration file to change the setting however you like. We can help guide you in this.

Summary

- Use HPCC AWS Cloud to provide an on demand HPC Cluster composed of the exact amount and type of compute resources you need for your research.
- Easy setup
- Easy operation
- Fully featured
- Billed directly by AWS

Questions

